# MapOmega Core



**Field case: Germano Mine (Samarco)**

MapOmega Tecnologia da Informação Ltda. · Technology Innovation Park (PIT) — São José dos Campos, SP, Brazil

Ver. 2026-09-10 · English edition

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# 1. About MapOmega

MapOmega builds technology to anticipate risks, prevent accidents, minimize losses and reveal hidden opportunities in operations. **MapOmega Core** is the first application of that mission, focused on mining.

Over 30 years of industrial methodology, validated in more than 30 steel mills, underpin the product — this isn't startup theory, it's hands-on practice from people who have been on the shop floor.

### BY THE NUMBERS

- 30+ years of industrial methodology
- 30+ steel mills validated
- 1 real-world mining case — Samarco

### ECOSYSTEM

MapOmega operates within the NEXUS ecosystem, the innovation hub of the Technology Innovation Park (PIT) in São José dos Campos, Brazil — one of the world's top five innovation environments, alongside Embraer, ITA, INPE and UNIFESP. This ecosystem is more than an address: it is the network of technical validation, acceleration and partnership behind every product in the MapOmega portfolio.

*Partners & credentials: PIT · SJC — NEXUS Growth — SEBRAE ES — Base27 — Ventiur — SAMARCO — Mining Hub.*

# 2. The problem — the risk is regulated, the monitoring isn't

Whole-body vibration, noise and heat — standards such as Brazil's NHO-09, NHO-01 and NR-15 (aligned with ISO 2631 for vibration) already define what must be measured, and how rigorously. In practice, most operations still measure it by campaign: a one-off measurement, one report a year, a technician with a borrowed dosimeter for a single shift.

**The risk is there every single shift. The data isn't.**

### REGULATORY TRIGGER

Stepped-up MSHA enforcement in the US and the requirement for continuous proof of exposure through eSocial/PGR in Brazil make the gap between campaigns ever more expensive to sustain — both in health risk and in legal exposure.

| | Today | MapOmega Core |
|---|---|---|
| **Measurement frequency** | 1× / year | Continuous |
| **Operator coverage** | Sample-based | Individual, 100% |
| **When the risk is found** | In the report | In real time |

# 3. Why wearables alone fall short — and how we solved it

Instead of instrumenting the machine, we instrument the person operating it. A wearable with embedded AI, read continuously — not one campaign a year.

### WHY A WEARABLE

- No installation
- No maintenance
- Individualized — per operator, not per piece of equipment
- Affordable
- Autonomous / *Wear-and-forget*

### THE CATCH

On its own, a wearable doesn't have the accuracy the standard requires — and that creates two problems.

False positive: false alarms stress Health & Safety teams and increase adoption friction.

False negative — the worse of the two: a real risk goes unreported. That is exactly what the standard exists to prevent, and it is a scenario no occupational health and safety team can accept in exchange for "convenience".

## Both problems make wearables economically promising but technically unviable — and the same holds, today, for very expensive direct, certified measurement sensors: technically viable, but economically unviable for continuous use.

### THE SOLUTION — INDIRECT MEASUREMENT, INFERRED BY AI

Continuous direct measurement — with a certified instrument, operator by operator, every shift — is economically unviable: it would mean replicating, at scale, the cost of a campaign dosimeter per operator, every day. Indirect measurement, inferred by an embedded AI model from affordable wearable sensors, is possible.

We proved it at the Germano Mine (Samarco): the model's inferences deviated by at most 0.039 m/s² from the reference standard measurement, with 84.3% accuracy. That is enough to deliver a priority-attention list, based on accumulated risk exposure, shift by shift — not a compliance report, but the data no annual campaign can deliver in the gap between one measurement and the next.

### TWO FORMATS, TODAY

The source can be any sensor — today we have two in the field.

| | Format 1 — Chest wearable + app | Format 2 — Integrated belt clip |
|---|---|---|
| **Sampling** | 200 Hz | 200 Hz |
| **Where inference runs** | In the operator's app (smartphone), via BLE | On the device itself (edge) |
| **Upload to cloud** | Wi-Fi/LTE, every 10 s (via app) | Wi-Fi/LTE, every 10 s (direct) |
| **App dependency** | Yes | No |

*200 Hz × 10 seconds, converted by AI in every 10 s window — the same model that reduces false positives and eliminates false negatives: 2,000 raw readings become 1 equivalent whole-body vibration measurement. In both formats, the data converges into the same JSON in the cloud — consolidated and delivered to the manager on the dashboard.*

### THE ANSWER

The result is simple: a list of the most exposed operators, updated every shift. Behind it, an AI model translates the sensor reading — a wearable today, any other data source tomorrow — into a reading aligned with the same thresholds the standard uses, reducing false positives and, above all, eliminating false negatives. The source may change. The list is what the occupational health and safety team receives.

### BEFORE YOU ASK: IS MAPOMEGA CORE A CERTIFIED INSTRUMENT?

No — and that isn't our role. Legal proof of compliance still requires the official survey, with a calibrated instrument — that doesn't change, and it never will. **MapOmega Core** does not compete with that.

Our role is a different one: to show Occupational Health who is most exposed day to day — data that a periodic campaign, by definition, does not deliver, because continuous monitoring with a certified instrument, operator by operator, every shift, is economically unviable. That is exactly where enforcement falls short today — not out of ill will, but because the annual-campaign model was never designed to be continuous.

**The law proves compliance once a year.**

**MapOmega Core shows exposure every single day.**

## 4. What we monitor today

We don't replace the official occupational hygiene survey — we reference the same thresholds it uses, to cover the gap between one certified measurement and the next.

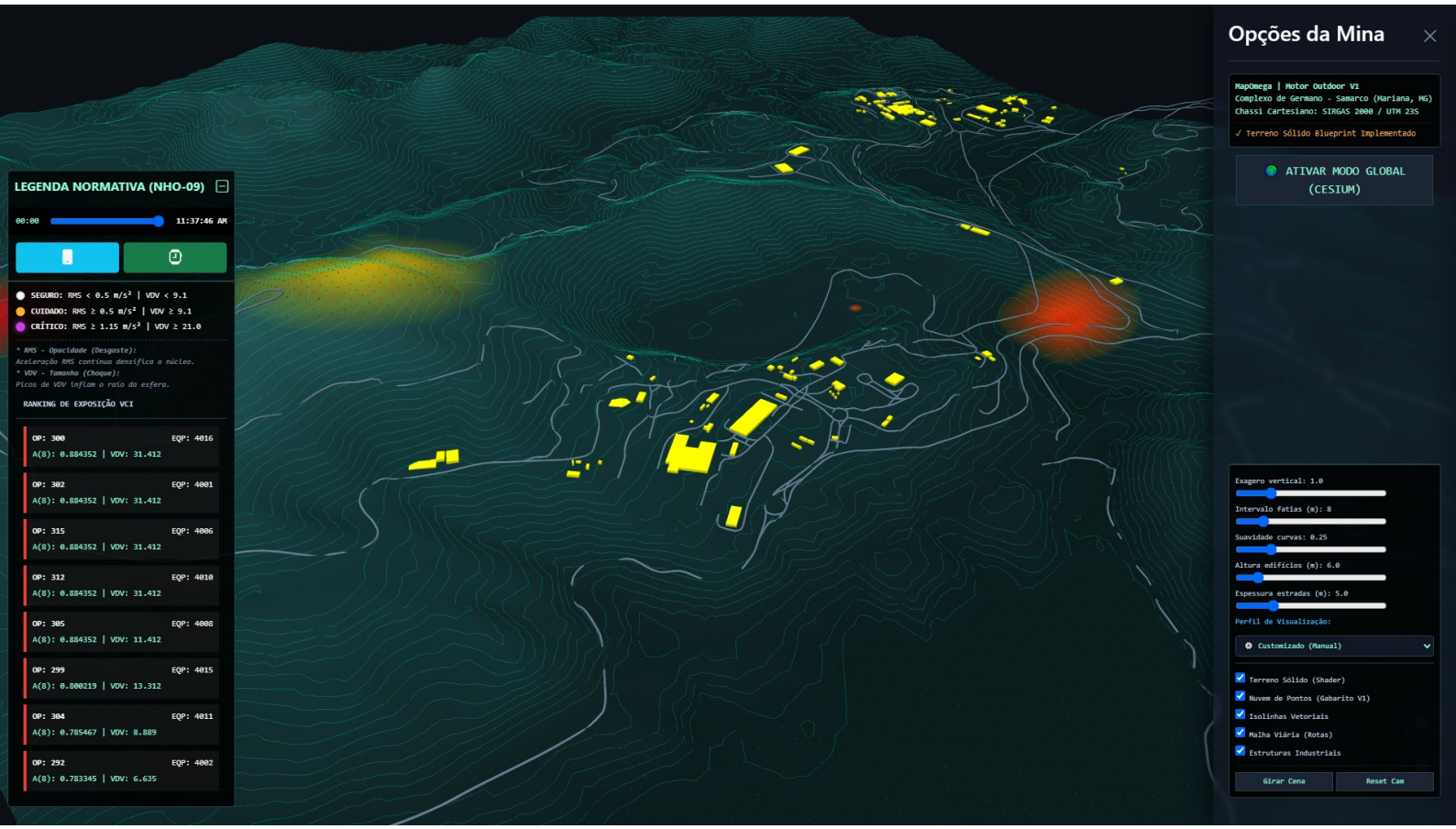
**MapOmega Core** was designed from day one as a multi-sensor architecture. Today, Whole-Body Vibration (WBV) is the front validated in the field, with a proof of concept running in production. Noise and Heat are part of the engineering roadmap — the platform and data protocol are already prepared to incorporate them — but combined sensing is still under development and should not be read as commercially available today.

| Quantity | Reference standard | What we measure | Status | TRL |
|---|---|---|---|---|
| **Whole-Body Vibration (WBV)** | NHO-09 / ISO 2631 | aren (A(8)), VDV and crest factor, by operator/equipment/route | **Field-validated** | 8–9 |
| **Noise** | NHO-01 | Estimated daily dose per operator | **Under development** | 3–4 |
| **Heat** | NR-15 | WBGT index, limit by metabolic rate | **Under development** | 3–4 |

*TRL = Technology Readiness Level (1–9). Vibration is validated in live operation; Noise and Heat are in hardware research and development, with no announced commercial date.*

## 5. Real-world case — Germano Mine (Samarco)

**MapOmega Core** was field-validated at the Germano Mine, operated by Samarco in Mariana, Minas Gerais, Brazil, focusing on Whole-Body Vibration (WBV) monitoring of heavy-equipment operators, according to NHO-09 criteria.



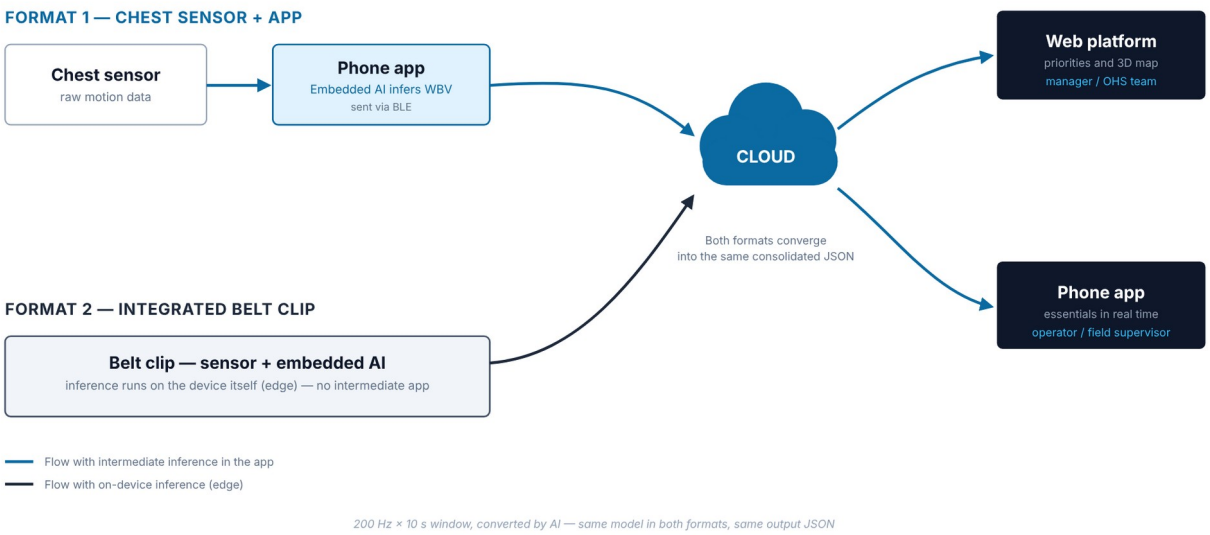

We built the 3D terrain model of the Germano Mine from real elevation data (QGIS survey, 30 m resolution), with a whole-body vibration dashboard running over the operation's real topography — an interpolated heatmap, updated continuously.

### WHAT THE CASE DEMONSTRATES

- AI inference validated in the field: deviations of at most 0.039 m/s² from the reference standard, with 84.3% accuracy
- Continuous exposure reading, shift by shift, without replacing the certified report
- Individualized prioritization — who is most exposed, right now, not the workstation average
- 3D spatial view of the mine, with exposure data overlaid

## 6. How it works — the platform

A single web platform of individualized priorities, with a 3D spatial map — fed independently by the two sensor formats presented in section 3, chosen according to the operation's priorities.




*Data flow: from the two sensor formats, converging in the cloud, to the web platform and the operator app*

The web platform holds the full configuration and the priorities by equipment, operator or location in the mine. In the app, the operator and the field supervisor get the essentials in real time — without depending on the central BI.

# 7. Technical sheet

## 7.1 Technology readiness

Format 1 (chest wearable + app) and Format 2 (integrated belt clip) are at different stages of maturity. The full technical sheet — with TRL detail per format, installation requirements and hardware specifications — is delivered at the PoC stage, together with the client's technical team.

## 7.2 Integration

- REST API to automatically link exposure to shift and operator
- Cloud hosting by default; on-premises can be evaluated after a technical assessment
- Platform designed to take in other data sources — camera, radar, in-cab sensor — as the operation requires

# 8. The cost of doing nothing

A lost-time case costs more than it seems. Automation, even more.

## 8.1 Cost of an unanticipated lost-time case

Each exposure-related lost-time case — heat, noise or vibration — carries direct and indirect costs that only surface after the harm is done. Reference figures for Brazil:

| | |
|---|---|
| Work-injury sick pay (INSS, social security) | **from day 16** |
| Full FGTS (severance fund) deposits during leave | **mandatory** |
| Job protection after return | **12 months** |
| Operator replacement / retraining | **weeks** |

| | |
|---|---|
| Higher FAP/RAT — work-accident insurance contribution | **up to 2×** |
| Risk of fines / shutdown (MSHA / NRs) | **variable** |

*FAP: an actual Brazilian social security mechanism (Law 10.666/03) — a company's accident history can raise its RAT rate by up to 100%, year after year, until it brings its claims rate down. Other estimates: studies of the cost of musculoskeletal disorders in Brazil point to ~US$ 2.2 billion in 2012–2016 for low back pain alone, with 79% of the cost coming from lost productivity — market reference figures, not contractual.*

### 8.2 The automation counterpoint

Autonomous/electric equipment solves the same exposure risk — by removing the operator from the cab.

# US$ 5–8M

*per unit — 240–400 t mining haul truck, market data (CAT / Vale)*

**MapOmega Core** addresses the same risk for a fraction of that investment — keeping the operator, the job and the operation as they are — and also generates the data that will justify, or safely postpone, the automation decision in the future.

*MapOmega's revenue unit is the "monitored life": the value delivered is measured by the reduction of unmanaged exposure per operator, shift by shift — not by the number of sensors sold.*

## 9. How we deploy

**STEP BY STEP**

- PoC — proof of concept, reduced scope, technical validation in the field
- Pilot — controlled expansion, with a real operating crew
- Contract — full scale, defined SLA, API integration

### 9.1 Quick sheet for the IT team

- Connectivity: Wi-Fi/LTE, transmission every 10 seconds
- Integration: REST API, automatic link to shift and operator
- Data: cloud hosting by default; on-premises after a technical assessment

## 10. Frequently asked questions

### MapOmega Core: a certified instrument?

No — and that isn't its role. Legal proof of compliance still requires the official survey, with a calibrated instrument. **MapOmega Core** covers the gap between one survey and the next, showing who is most exposed day to day.

### Does it replace certified measurement?

No. It shortens the blind interval between reports, with continuous readings aligned to the standard's thresholds — but the legal proof of compliance remains the official survey.

### Where is the data stored?

In the cloud, by default. On-premises hosting can be evaluated after a technical assessment with your IT team.

### Does it integrate with the timekeeping system?

Yes, via REST API — exposure is automatically linked to shift and operator.

### Does it only work with these two sensors?

Today we have two formats in the field — a chest sensor with an app, and a standalone belt clip. The platform was designed to take in other sources as the operation requires.

## 11. Talk to the team

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